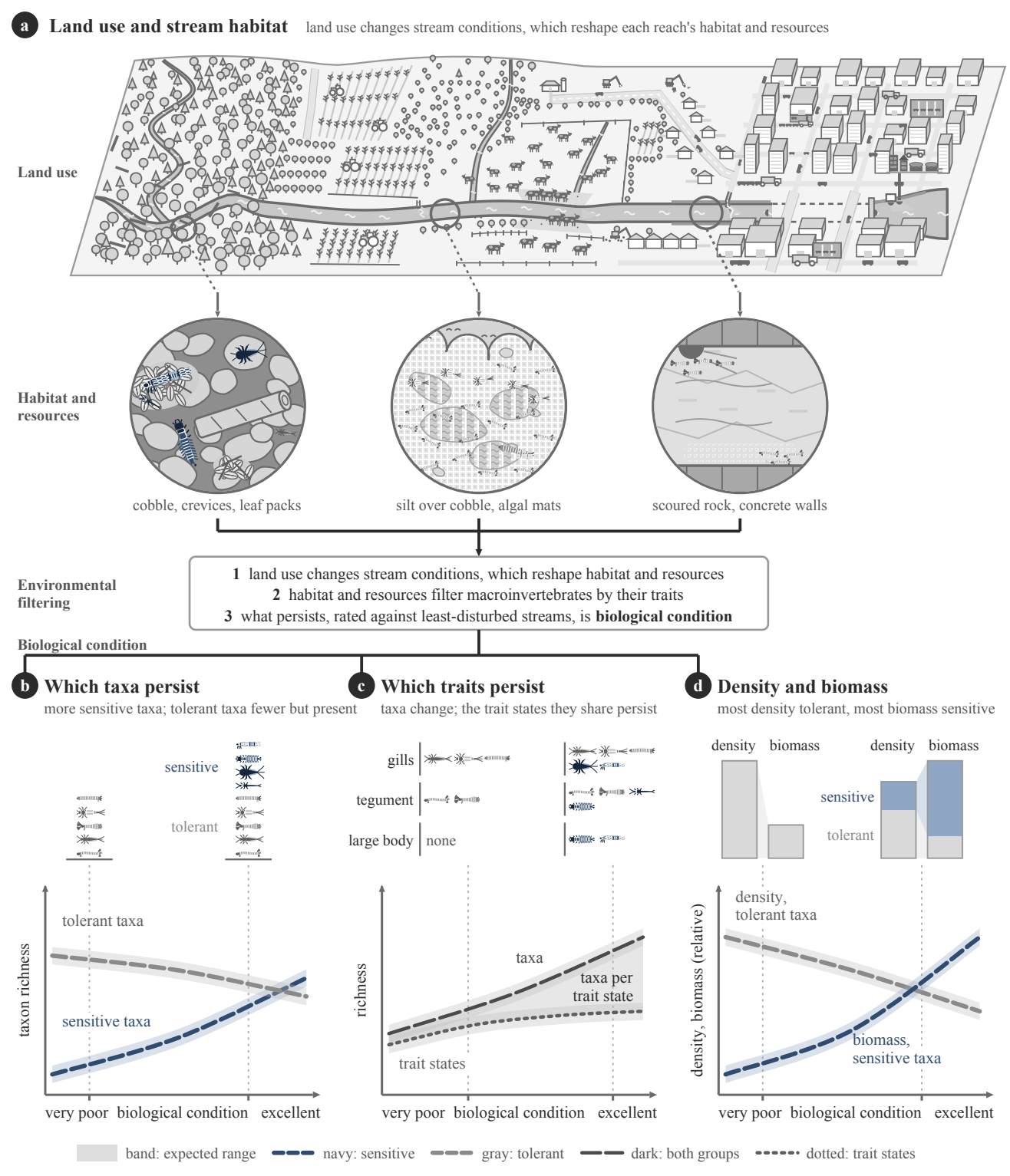

a Land use and stream habitat
land use changes stream conditions, which reshape each reach's habitat and resources
Land use
Habitat and resources
cobble, crevices, leaf packs
silt over cobble, algal mats
scoured rock, concrete walls
Environmental filtering
1 land use changes stream conditions, which reshape habitat and resources
2 habitat and resources filter macroinvertebrates by their traits
3 what persists, rated against least-disturbed streams, is biological condition
Biological condition
b Which taxa persist
more sensitive taxa; tolerant taxa fewer but present
sensitive
tolerant
tolerant taxa
sensitive taxa
taxon richness
very poor
biological condition
excellent
c Which traits persist
taxa change; the trait states they share persist
gills
tegument
large body
none
taxa
taxa per trait state
trait states
richness
very poor
biological condition
excellent
d Density and biomass
most density tolerant, most biomass sensitive
density
biomass
density
biomass
sensitive
tolerant
density, tolerant taxa
biomass, sensitive taxa
density, biomass (relative)
very poor
biological condition
excellent
band: expected range
navy: sensitive
gray: tolerant
dark: both groups
dotted: trait states